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universidade de aveiro
departamento de eletrónica,
telecomunicações e informática

Observability on NopCommerce

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Product Search Flow

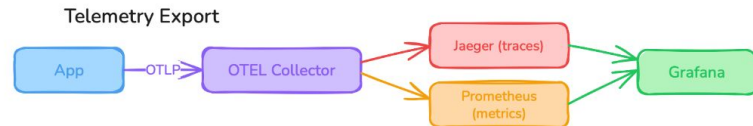
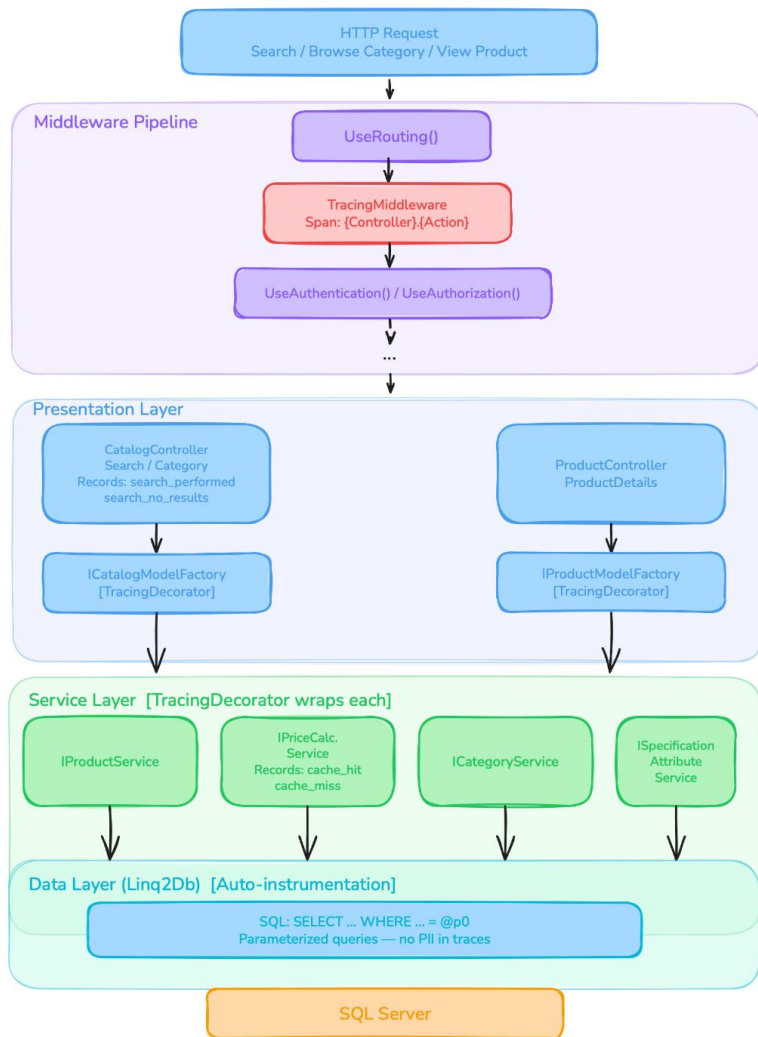
Customer searches and views a product

Goal: Add OpenTelemetry Tracing & metrics

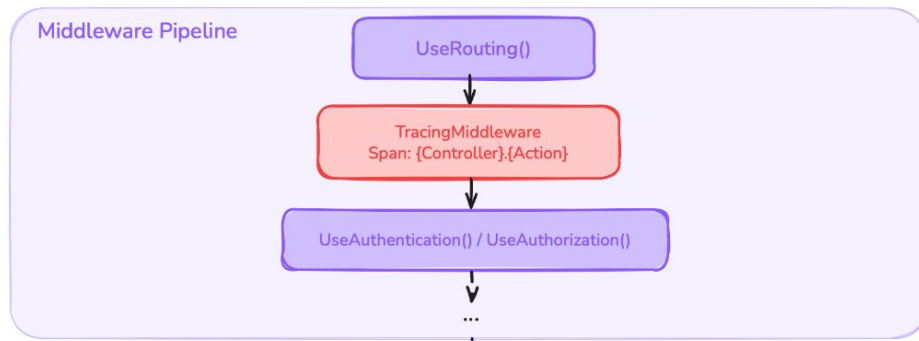
Constraint: Minimize changes to business logic

Services involved:

- Catalogue
- Product
- Pricing

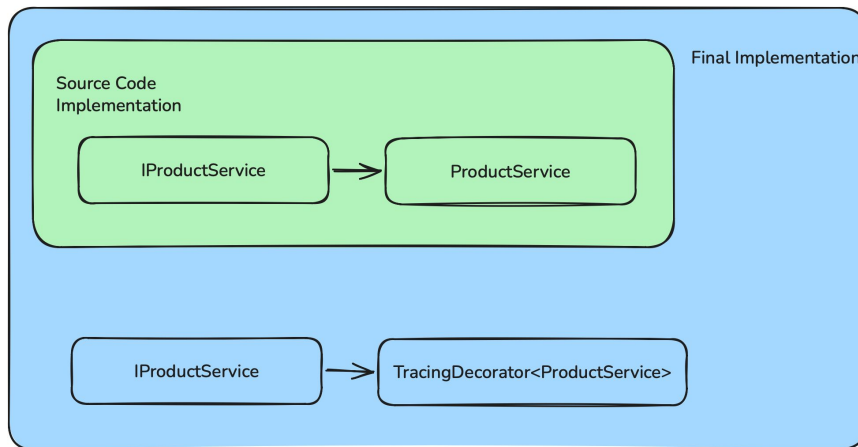
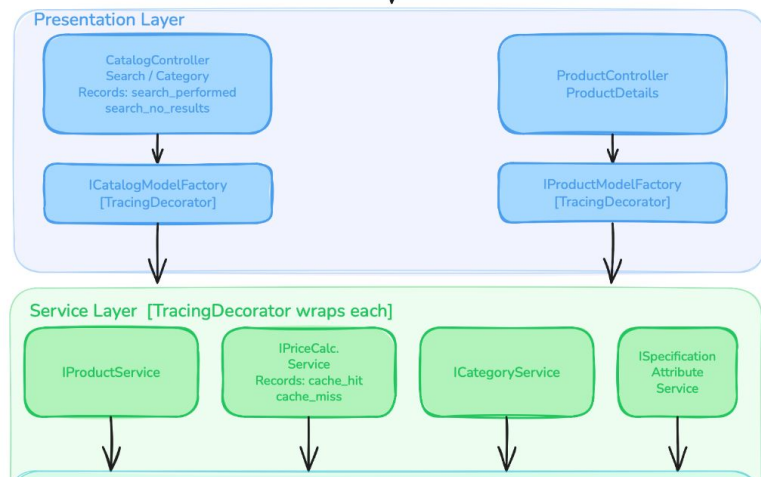


Tracing – Middleware



- Added new layer in middleware
- Tracks every request to the app
- Creates spans for every controller endpoint
- No controller code modified

Tracing – Services and Factories



- DispatchProxy wraps interface at DI registration
- Every method call -> new span
- Zero business logic modified

Tracing – OpenTelemetry Auto Instrumentation

- Automatically traces SQL Queries
- Automatically traces HTTP calls

Sensitive Data

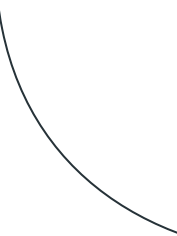
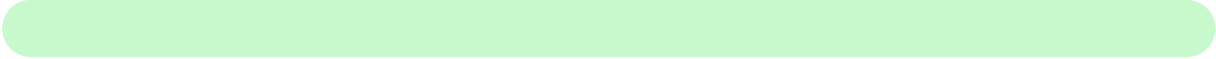
Included: IDS, booleans, enums

Excluded: strings, objects

Metrics

- Search_performed_total
 - Search_no_results_total
 - Pricing_cache_hit
 - Pricing_cache_miss
 - **Search Pipeline Health**
 - **Pricing Cache Health**
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- No-results ratio rising → broken search
- Cache hit rate dropping while search volume rises → load pressure hitting the database
- Cache hit rate stable while search volume rises → system absorbing the traffic



Demo



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